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The O-RAN Whitepaper 2026

RAN Automation with SMO and rApps

Recently, the O-RAN automation market has been shifting toward the SMO/Non-RT RIC solutions. At the same time, due to the complexity of current systems, operators demand dynamic, intelligent, context-aware network automation. According to the O-RAN architecture, this can be realized by dedicated rApps residing in SMO/Non-RT RIC, typically utilizing AI/ML tools. This whitepaper provides an overview of the RAN automation with SMO and rApps. First, the SMO/Non-RT RIC architecture in O-RAN is introduced. It is followed by discussions on two promising use cases of RAN automation: traffic steering and beamforming optimization. The whitepaper addresses both general aspects and the design of related rApps, along with performance results.

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Executive Summary

O-RAN standard development continues driving innovation in intelligent RAN management. The current telecom market is moving closer to adopting the Service Management and Orchestration (SMO)/Non Real-Time RAN Intelligent Controller (Non-RT RIC) than Near Real-Time (Near-RT) RIC. At the same time, Mobile Network Operators (MNOs) are pushing towards intelligent automations to replace the state-of-the-art threshold-based approach to network management. This goal can be effectively realized by introducing third-party RAN automation algorithms known as rApps. This specifically takes into account that Network Equipment Vendors (NEPs) are announcing dedicated automation platforms like Ericsson EIAP or Amdocs Cognitive RAN Automation. By utilizing AI/ML tools, modern rApps are expected to provide MNOs with RAN automation algorithms, helping with Radio Resource Management (RRM) tasks.

This whitepaper provides a technical discussion of RAN automation using rApps. After introducing the SMO/Non-RT RIC in the context of O-RAN architecture, the discussion focuses on two practical use cases: traffic steering and beamforming optimization.

The first chapter provides an overview of the functional architecture of SMO and Non-RT, as defined by the O-RAN ALLIANCE specifications, and introduces the concept of rApps. It serves as a starting point for more details provided in the subsequent parts.

We continue the discussion on traffic steering design within Non-RT RIC. Considering a typical handover management, our initial design was an xApp implementation for Near-RT RIC, while recently, we designed its rApp version. Chapter 2.0 provides a comprehensive comparison of Non-RT and Near-RT RICs' capabilities for traffic steering features.

Following this, Chapter 3.0 introduces the Traffic Steering rApp (TS-rApp). First, we provide a detailed description of its design. This includes: algorithm, input/output data, and signaling flow. Subsequently, the discussion continues with an actual implementation on the O-RAN Software Community-RIC (OSC-RIC) connected to the VIAVI network emulator under the SMaRT-5G initiative.

The next two chapters touch upon another aspect of 5G and 6G networks management, namely, Massive MIMO (M-MIMO), and the related Beamforming Optimization (BO). To this end, in Chapter 4.0, we discuss how to perform beamforming optimization in an M-MIMO system utilizing a predefined set of beams, i.e., Grid of Beams (GoB). This is reflected by the four identified BO use cases. Chapter 5.0 follows with an example design and implementation of the BO-rApp, highlighting the use of Network Digital Twin (NDT). This is supplemented by the results showcasing the benefits of the BO-rApp in a scenario of a large social event, resulting in users' mobility patterns different from a typical day-to-day network operation.

The whitepaper ends with a summary and conclusions section, along with a glossary of terms.

1.0

Introduction to SMO/Non-RT RIC

The goal of Non-RT RIC is to support intelligent RAN automation by providing policy-based guidance, Configuration Management (CM), and data collection for network optimization. Non-RT RIC resides in the management plane (specifically in SMO) and hosts third-party algorithms packaged as rApps. They perform their RAN automation tasks on a timescale of tens of seconds to tens of minutes.

In this chapter, we provide the architecture of the Non-RT RIC and discuss example rApps.

1.1 O-RAN Non-RT RIC

The functionality of the Non-RT RIC includes CM, device management, fault management, Performance Management (PM), and lifecycle management for all network elements within the O-RAN architecture. It is similar to the Element Management System in the traditional networks. All RAN elements can be self-configured by the Non-RT RIC, reducing the need for manual intervention. By providing timely insights into network operations, MNOs may use Non-RT RIC to better understand and optimize the network by applying pre-determined service and policy parameters. Its functionality is internal to the SMO in the O-RAN architecture and provides the A1 interface to the Near-RT RIC. Non-RT RIC can use data analytics and AI/ML training/inference to determine RAN optimization actions for which it can leverage SMO services, such as data collection and provisioning. Trained models produced in the Non-RT RIC are either directly used for RAN optimization over the O1 interface terminated in SMO or distributed to the Near-RT RIC for runtime execution. Network slicing, security, role-based access control, and RAN sharing are example aspects enabled by the combined controller functions, namely, Near-RT and Non-RT [1]. At the same time, Non-RT RIC hosts standalone, third-party RAN automation algorithms packaged as rApps, and provides them with services over the R1 interface. The related use cases can cover, e.g., Traffic Steering (TS), energy savings, or Beamforming Optimization (BO).

1.2 O-RAN Non-RT RIC Architecture

Fig. 1.2-1 shows the internals of the SMO framework, including the Non-RT RIC. Functions inherent to the Non-RT RIC itself are used for managing the rApps, which are external to the Non-RT RIC framework, accessible through an open Application Programming Interface (API), using the R1 interface. Another set of functions includes those that create the data to be transmitted over the A1 interface, namely: A1 policy functions, A1 enrichment information functions, and A1 ML functions.

There are also parts of the SMO framework that are out of Non-RT RIC scope, namely those related to the O1/O2 termination or SMO framework functions, e.g., for network slicing lifecycle management. Those are inherent to the SMO framework.

Finally, there is a part of functionality whose implementation is flexible. The related functions could be part of Non-RT RIC, or could be external to Non-RT RIC and sit in the SMO. The most prominent example here is AI/ML workflow functions. In this context, AI/ML could be either part of Non-RT RIC or external, providing information to the SMO service exposure functions.

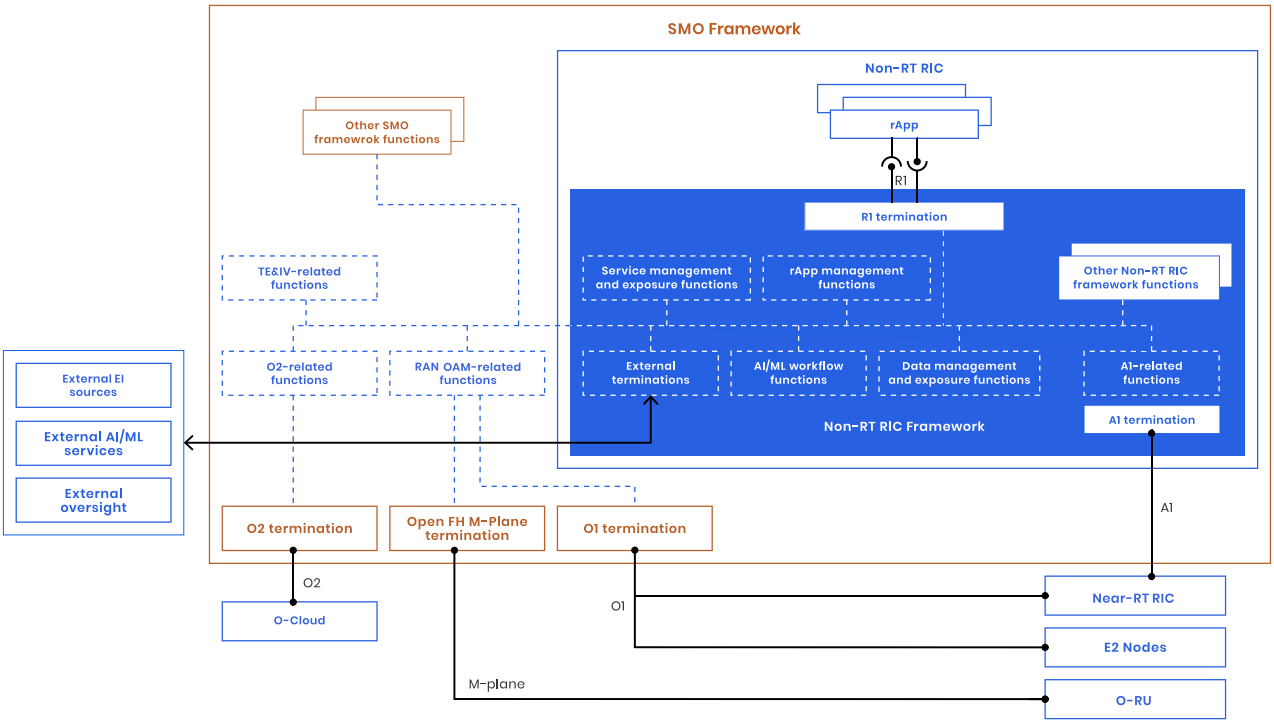


Fig. 1.2-1. Detailed functional architecture (based on [1])

1.3 rApp Examples

rApps are applications designed to run on Non-RT RIC, providing additional value-added services to help create policies that the Non-RT RIC framework delivers to Near-RT RIC over the A1 interface, or to directly perform RAN automation tasks executed by SMO over the O1 interface. Fig. 1.3-1 shows Non-RT RIC with two example rApps, namely Traffic Steering rApp (TS-rApp) and Beamforming Optimization rApp (BO-rApp), connected to the Non-RT RIC framework via R1 interface.

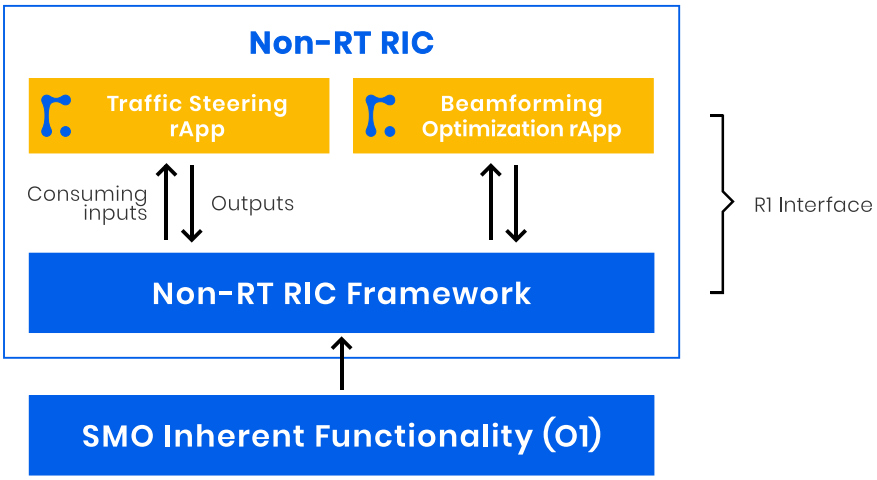


Fig. 1.3-1. Example rApps

Traffic Steering rApp aims to perform Load Balancing (LB) features. For this purpose, it utilizes O1 measurements of cell load, along with carrier frequency and bandwidth configuration of individual cells. The internal logic of TS-rApp analyzes these data and modifies a Cell's Individual Offset (CIO) over the O1 interface. These affect the user-to-cell association and enable active off-loading of traffic from congested cells to underutilized cells. TS-rApp:

» Consumes: O1 measurements of cell load, regarding actual capacity utilization for a cell site over time, along with the cell's carrier frequency and bandwidth;

» Outputs: CIO, which affects the user-to-cell association, enabling long-time-scale LB.

Beamforming Optimization rApp aims to adjust the direction of beams in an M-MIMO system to improve network coverage and SINR. For its decision process, BO-rApp uses the O1 interface to obtain network configuration like cell, carrier, and sector parameters, along with directions of individual beams. Based on O1 measurements like Reference Signal Received Power (RSRP) or Signal-to-Interference plus Noise Ratio (SINR), potentially augmented with location information provided by SMO from an external source, the BO-rApp decides on the adjustment of beam directions and monitors the resultant performance. BO-rApp:

» Consumes: O1 network configuration like cell, carrier, sector parameters, parameters of individual beams (width, azimuth, elevation), and measurements of RSRP and SINR;

» Outputs: modified beamforming parameters including digital tilt and azimuth, aiming to improve network coverage or reduce interference.

Both rApps can be equipped with an AI/ML model to improve their performance, or utilize other SMO/Non-RT RIC services such as NDT, for initial verification of their decisions in a safe environment.

Throughout the next chapters, we will go into detailed discussions, practical examples, and most importantly, results related to TS and BO. Chapters 2.0 and 3.0 cover the design and implementation of TS-rApp, a result of migration from TS-xApp. Following this, Chapters 4.0 and 5.0, discussing BO-rApp, are focused on the design of the solution targeted from the beginning as rApp.

2.0 Traffic Steering Migration to Non-RT RIC

TS-xApp has proven highly effective for near-real-time network optimization [2][3][4][5]. However, current networks move closer to adopting the SMO/Non-RT RIC, and operators are considering rApps deployments before xApps [6]. To address this, we have rethought how certain features of the TS-xApp can be migrated from an xApp to an rApp. This redesign encompasses a different timescale, interfaces, and algorithmic approach.

This chapter extends the general discussion on the architecture of SMO/Non-RT RIC with a detailed analysis of the traffic steering use case in the context of O-RAN architecture.

2.1 Traffic Steering Functionalities

TS is a fundamental RRM function in mobile networks that determines which cell or base station (BS) a user equipment (UE) connects to. It can involve decisions for single or multiple connections, often using advanced features like Dual Connectivity or Carrier Aggregation to meet diverse user and application needs, while optimizing the use of available radio resources. Modern TS goes beyond simple, uniform rules. For example, the Rimedo Labs' TS-xApp integrated with different RIC platforms offers: service-based TS, A1 interface integration, cooperation with energy saving functions, and LB features. Ultimately, the TS is a multidimensional optimization process essential for delivering the best user experience in today's complex, heterogeneous mobile networks.

Precise control of user allocation to cells requires low latency, so the default and most appropriate choice is to place the TS algorithm in the xApp form. However, considering the operators' current preferences and the stages of RIC implementation in response to market needs, let us see how TS can be implemented differently. More specifically, to what extent the features of TS within Near-RT RIC available over the E2 interface could be translated to the O1 interface and Non-RT RIC.

2.2 Feasibility of E2 and O1 Interfaces for Traffic Steering

The key determinant for the xApp-rApp migration is the availability of input data and the possibility of controlling the RAN. Let's now look at the differences between the E2 interface (used by TS-**xApp**) and the O1 interface (used by TS-**rApp**), summarized in Table 2.2-1 (data reporting) and Table 2.2-2 (RAN control capabilities). Further, Table 2.2-3 highlights the main differences between TS-xApp and TS-rApp.

For monitoring RAN performance, both O1 and E2 interfaces rely on the PM metrics defined by 3GPP in TS 28.552 [7]. However, as can be seen in Table 2.2-1, there are two major differences:

» **Scope:** While the O1 interface can expose the PM data, for E2 Node, cell, network slices, and 5QI, the E2 interface offers flexibility by extending the scope of some PM data even to the UE-level.

» **Report Interval:** Although, in theory, the Non-RT RIC control loop might be as low as tens of seconds, in practice, the PM data exposed over the O1 interface are reported within intervals of minutes or even tens of minutes [8]. Instead, over the E2 interface, the reporting interval can be as low as tens or hundreds of milliseconds. This enables fast reaction to the changes in RAN performance.

In addition, the E2 interface provides a RAN Control (RC) Service Model (SM) [9], which can be used to copy messages from the RAN. For example, in the TS case, this could be measurement reports containing the RSRP of individual UEs. This is not possible over the O1 interface.

For obtaining the configuration of the cells, the O1 interface CM relies on Information Object Class (IOC) as specified by 3GPP TS 28.541 [10]. Some of these IOCs are exposed over the E2 interface through the Cell Configuration and Control (CCC) SM (e.g., O-CES Management Function for cell on/off switching). Based on our experience, in both cases, notifications to the xApp or rApp are sent upon the configuration change, and have a similar delay.

Service	Scope	Report Interval	Reported Data
O1 – PM	PLMN ID, E2Node, Cell, Slice, 5QI	Order of **minutes** (cyclic)	PMs according to 3GPP TS 28.552
O1 – CM	gNB, O-CU, O-DU	On configuration change	IOC, according to 3GPP TS 28.541
E2SM – RC	E2Node, Cell, Slice, 5QI, **UE**	Order of milliseconds (on configuration change /message copy)	RAN Control data according to the REPORT Service style from ORAN.WG3.E2SM-RC
E2SM – KPM	Plmn ID, E2Node, Cell, Slice, 5QI, **UE**	Order of milliseconds	PMs according to O-RAN.WG3.E2SM-KPM (based on 3GPP TS 28.552)
E2SM-CCC	E2Node, Cell, O-CU, O-DU	Order of milliseconds (on configuration change)	Cell Configuration according to O-RAN.WG3.E2SM-CCC

Table 2.2-1. Monitoring data available through E2 and O1 interfaces

As mentioned above, we identified differences between the O1 and E2 interfaces concerning data reporting. On the other hand, the means to control the RAN are crucial for the xApps and rApps design. To this end, there are two major differences (see Table 2.2-2):

» **Controlled parameter:** The E2 interface provides two major approaches to influence the RAN behavior in the context of TS. One is a direct control of the handover command. An alternative is to define the policies, to create flexible rules when a handover should be triggered, e.g., by setting proper A1-A6 thresholds [11]. In contrast, using the O1 interface, it is possible to control only the CIO. This reduces the flexibility and potential use cases for performing LB.

» **Scope:** The E2 interface offers a much more flexible scope of control actions. For example, both the handover commands and the policies can be defined for a cell, network slice, 5QI, or even for an individual UE. In contrast, the O1 interface allows only the CIO between a pair of neighboring cells to be defined.

Service	Controlled Parameter	Scope	Use Case
O1 – PM	CIO	Cell	LB (only long-term due to PM report delay)
E2SM – RC (CONTROL)	Handover command	Cell, Slice, 5QI, **UE**	LB, Service-Based TS, UE offloading for cell shutdown
E2SM – RC (POLICY)	A1, A2, ..., A6 thresholds	Cell, Slice, 5QI, **UE**	LB, Service-Based TS, UE offloading for cell shutdown

Table 2.2-2. Control and configuration possibilities available through E2 and O1 interfaces

Table 2.2-3 summarizes the main differences between the xApp and rApp variants of the TS algorithm implementation.

Aspect	xApp (Near-RT RIC)	rApp (Non-RT RIC)
Timescale	10ms – 1s	≥1s (typically minutes-hours)
Interface	E2 (direct RAN control)	O1 (config) + A1 (policy)
Control	Direct UE-to-cell decisions	CM
Algorithms	Real-time A3 events, RSRP handling	Offset for handovers
Deployment	Near-RT RIC	Non-RT RIC

Table 2.2-3. Comparison of TS-xApp and TS-rApp

2.3 Traffic Steering Integration Options

Taking into account differences between interfaces for data reporting and RAN control, Fig. 2.3-1 presents three approaches to the TS algorithm deployment in the network.

First, the TS function can be used as an xApp in Near-RT RIC (see the left part of Fig. 2.3-1). In this option, it operates by continuously monitoring real-time metrics from the RAN, and through dynamic reassignment of user traffic between cells, bands, or network slices based on factors such as radio conditions, cell load, user profiles, and Key Performance Indicator (KPI) triggers, aiming to maximize throughput, ensure efficient resource utilization, and balance load across cells. As indicated in the previous section, the E2 interface can expose PM data in short intervals and within a narrow scope, e.g., per UE, or per 5QI. Moreover, the RAN control capabilities allow control of individual handovers. This makes Near-RT RIC a straightforward choice for TS-xApp, especially for scenarios requiring rapid optimization, such as high user mobility or fluctuating traffic patterns in heterogeneous networks.

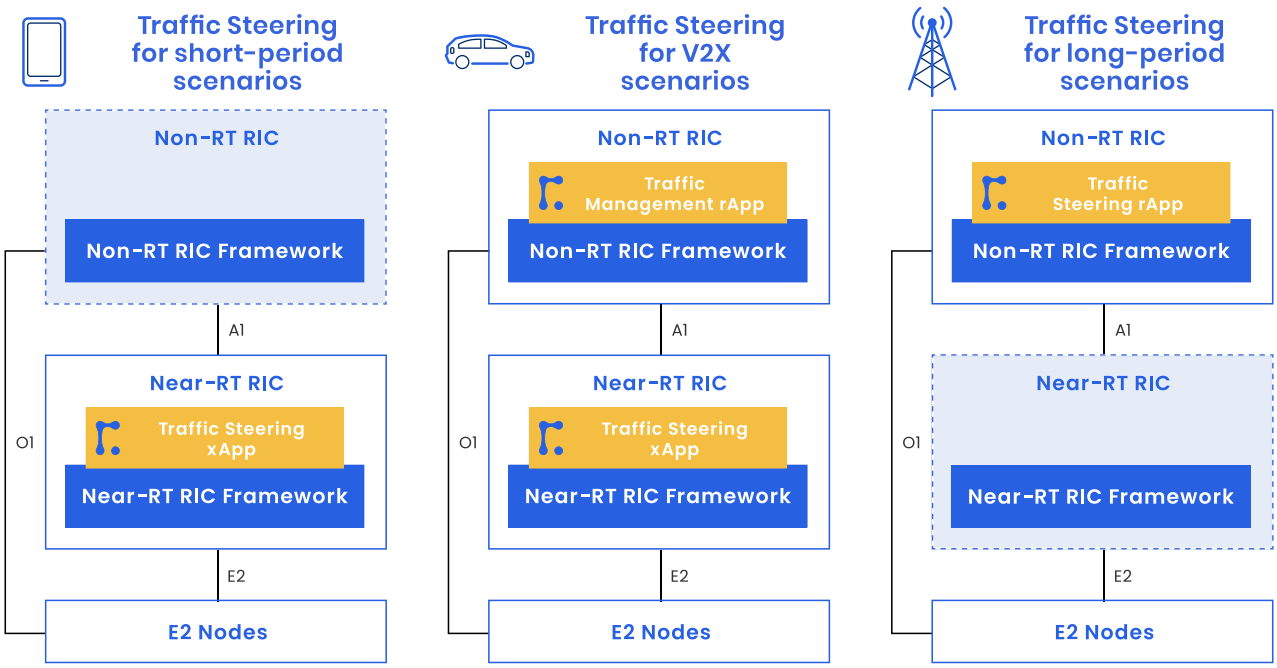


Fig. 2.3-1. TS integration variants

The broadest and most universal TS operation can be achieved by the joint usage of the TS-xApp with another rApp (see the central part of Fig. 2.3-1). In such an approach, the TS-xApp makes an immediate UE-related decision, and the rApp, through policies, optimizes TS-xApp operation to match a large-scale objective. This tandem uses the handover control offered by the E2 interface and the policy mechanism available through the A1 interface. In such a setup, the TS-xApp is cooperating with other applications. This is possible, due to the A1 policies it can receive from various rApps (e.g., Traffic Management rApp, or Energy Saving rApp [12], [13]). Such an app-tandem in the form of TS-xApp and Traffic Management rApp (TM-rApp) could be used for various advanced scenarios. As an example, TM-rApp analyzes the network PMs delivered through the O1 interface along with some context-information from external sources and formulates an A1 policy to change cell preferences for UEs. However, the exact execution of TS decisions is done by TS-xApp in Near-RT RIC. In [12], we considered such a TM-rApp and TS-xApp tandem, where TS-xApp operation is optimized by A1 policies formulated by TM-rApp based on information from an external source, e.g., a traffic accident detected in the network in a V2X scenario. The TM-rApp and TS-xApp tandem aims to ensure sufficient radio resources for the first responders and emergency services.

Finally, (as shown on the right part of Fig. 2.3-1), using TS as an rApp focuses on higher-level network optimization within the Non-RT RIC, operating on longer timescales, namely, minutes to hours. It steers the network by directly adjusting configurations through the O1 interface, rather than issuing policies via the A1 interface. As mentioned in the previous section, the O1 interface is characterized by long report intervals (e.g., a few minutes). Thus, the TS-rApp can analyze aggregated network data and PMs over a longer timescale. Moreover, its control capabilities are limited to modifying certain thresholds per cell (e.g., CIO), instead of triggering the explicit handover for a given UE. Therefore, in this case, the TS-rApp using O1 indirectly influences UE association by modifying cell parameters. The O1-driven approach is suitable for gradual, system-wide adaptations, but does not provide rapid, per-UE responsiveness found in near-RT control. As a result, TS as an rApp is best suited for scenarios focused on long-term network operational efficiency, like LB of the network traffic. The next chapter of this whitepaper discusses the implementation of such an rApp.

3.0 Traffic Steering rApp: Design & Results

The SMaRT-5G initiative, involving industry partners like Rimedo Labs, TietoEvry, VIAVI, Intel Labs, Aether, and Rakuten, did a Proof-of-Concept (PoC) with SMO and Non-RT RIC hosting rApps that operate solely via the O1 interface. We focused on migrating the TS function, originally designed as an xApp, to an rApp [14]. The joint activity is focused on developing the TS-rApp for 4G/5G heterogeneous RANs, leveraging an open-source O-RAN architecture.

While Chapter 2.0 covered the key differences between the implementation of TS features in Near-RT RIC and Non-RT RIC, Chapter 3.0 focuses on the actual design of TS-rApp.

3.1 Traffic Steering rApp Algorithm Design

Considering all the discussions in Chapter 2.0, our implementation of the TS-rApp enables an LB feature. As LB focuses on long-term network optimization, the O1 interface is used for monitoring purposes by observing aggregated PMs (on a minute-level scale). Using O1 CM, the effective cell coverage area (regarding the handover region) can be adjusted by setting the CIO.

The typical LB operation in a heterogeneous network is visualized in Fig. 3.1-1. When the coverage cell is overloaded, the LB feature moves some users to the capacity cell. Increasing the CIO of the capacity cell extends its effective coverage area, which results in some users moving from the coverage cell.

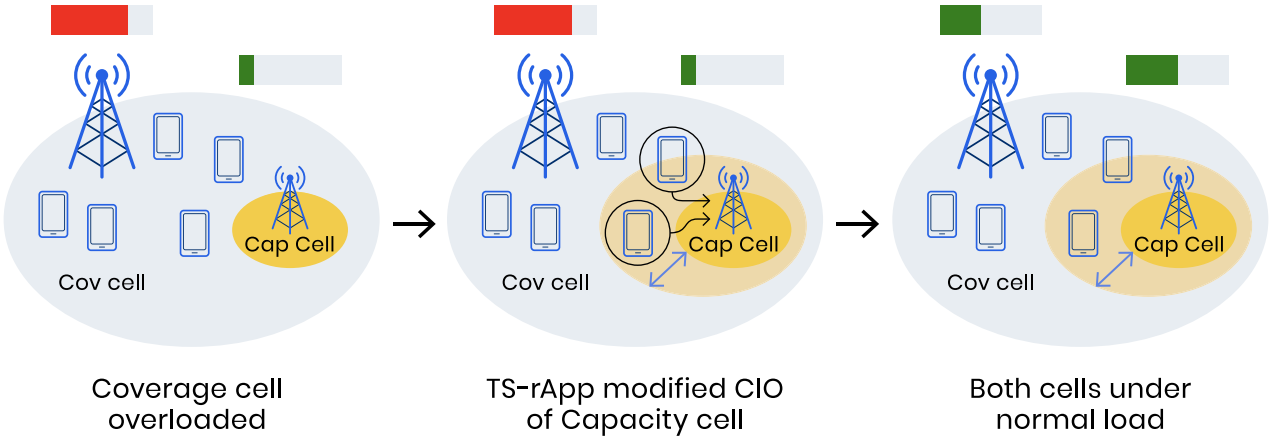


Fig. 3.1-1. The load-balancing feature of TS

Now that the general concept is known, let's move to the TS-rApp algorithm operation as shown in Fig. 3.1-2.

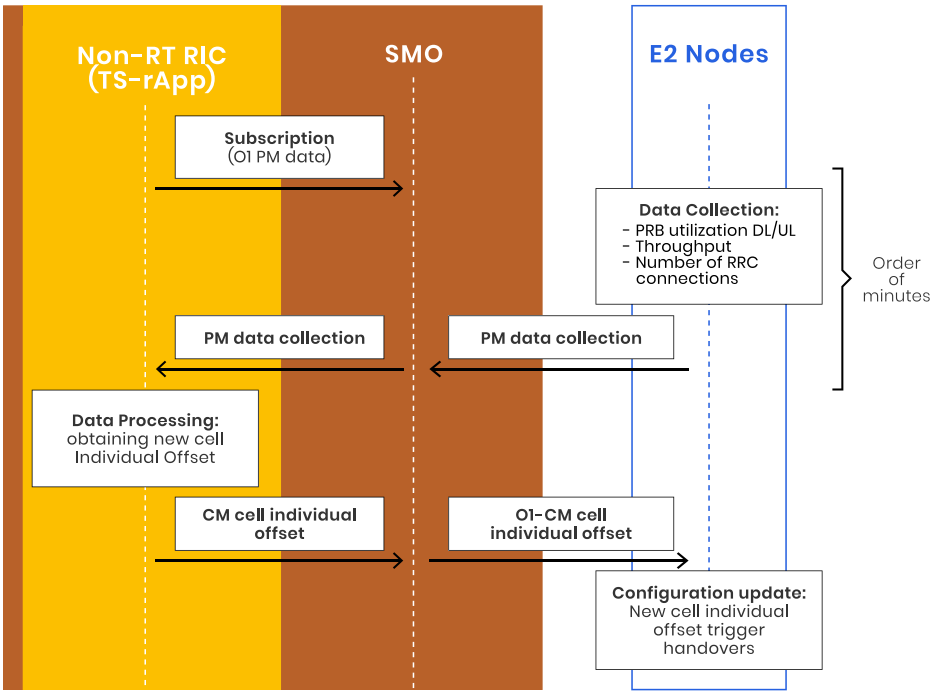


Fig. 3.1-2 Simplified signaling for TS rApp operation

TS-rApp starts with the subscription to the relevant O1 PM. In this case, the requested information is:

» Cell configuration available through CM [10], including

- arfcnDL/UL (for 5G), earfcnDL/UL (for 4G)
- bsChannelBwDL/UL (for 5G)

» Cell load available through PM [7], including

- DL/UL Total PRB Usage (RRU.PrbDl/Ul)

When the indicated metrics are collected, they are transmitted through SMO to the TS-rApp. Next, the internal TS algorithm calculates a new set of CIOs to match objectives (in this case, to balance load among cells). Using O1 CM, the following cell configuration can be set:

» TS-rApp controls indirect handovers through the CIO using CM [10], including

- rsrpOffsetSSB/CSI-RS
- rsrqOffsetSSB/CSI-RS
- sinrOffsetSSB/CSI-RS

Such a set of CIO values is sent to SMO and subsequently to relevant cells. Finally, the CIO is updated, which, in fact, changes handover regions and balances load in the network.

To realize the abovementioned idea, the adapted Rimedo Labs TS algorithm was integrated into the OSC Non-RT RIC as an rApp. Its operation was tested using the VIAVI network emulator. The simplified architecture of the solution is presented in Fig. 3.1-3.

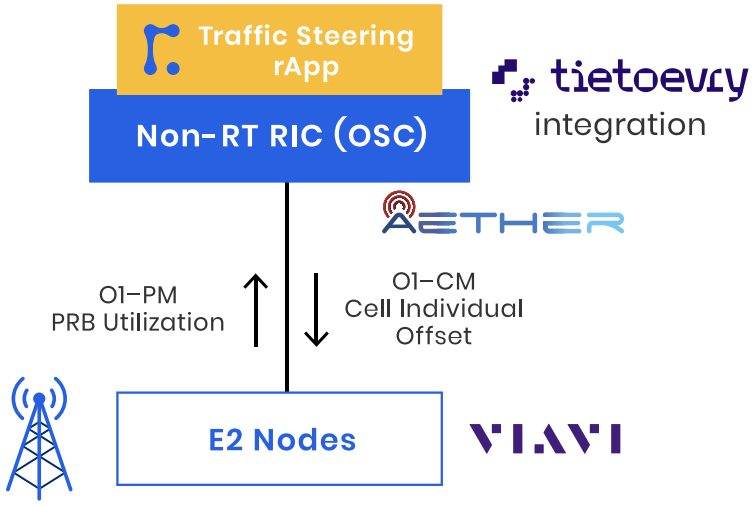


Fig. 3.1-3. Simplified architecture for the TS use case

3.2 Traffic Steering rApp Results

To verify the TS-rApp implementation, the following network setup was considered:

» Network topology: 17 cells, 80 UEs

» Carrier frequencies: 3.6 GHz for coverage cells and 28 GHz for capacity cells.

» Simulation duration: 40 minutes of the network operation.

LB improvement is visible in Fig. 3.2-1. Network traffic load (represented as PRB usage) from the highly loaded cells (IDs: 4, 6, 14) is offloaded to other neighboring cells.

Capacity cells typically use higher carrier frequencies; thus, larger path loss results in fewer UEs associated with these cells in the cell selection process. Correctly set, the CIO can prioritize capacity cell selection. The traffic offload operation (from coverage to capacity cells) lowers the RSRP of UEs, as some of them connect to not the strongest cell. However, as a result of such a transition, more resources are available for those UEs, thus not only is the high path-loss compensated, but also higher throughput is achieved for them.

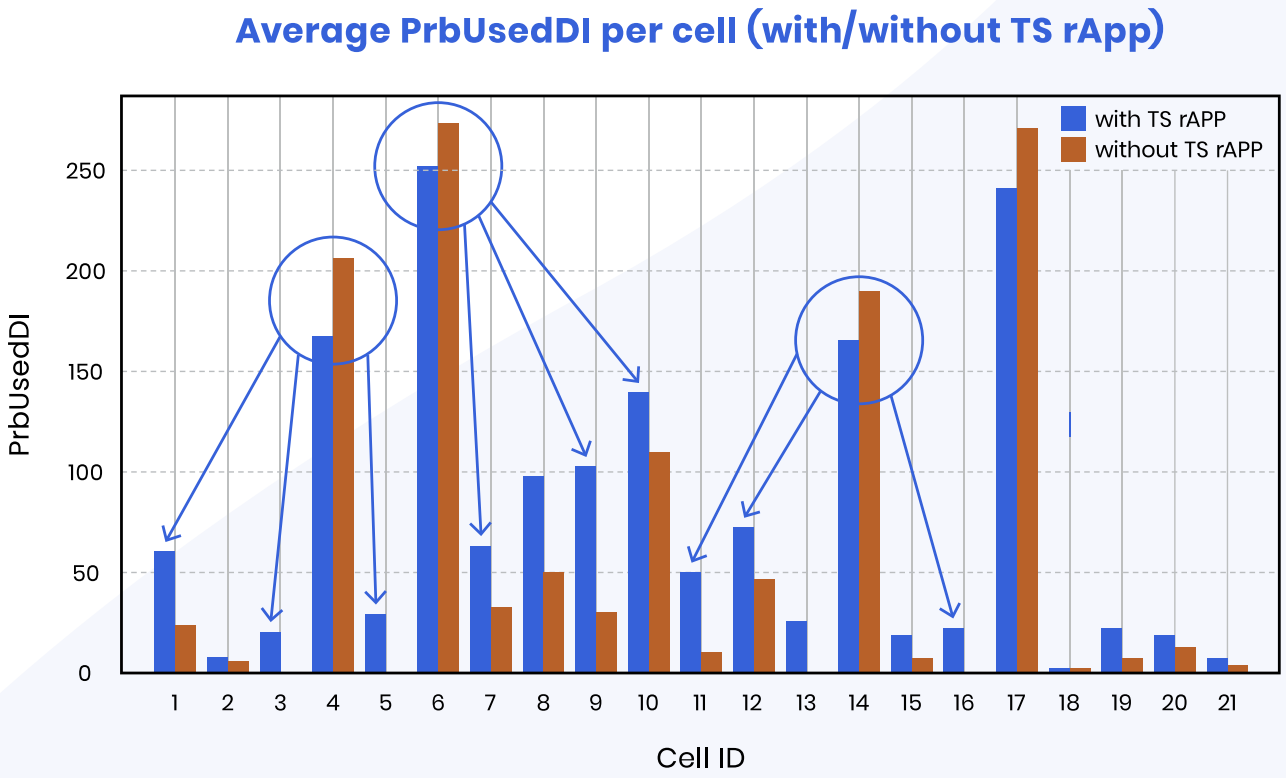


Fig. 3.2-1. Average number of usecd PRBs per cell in DL (with TS-rApp – blue, without TS-rApp – orange)

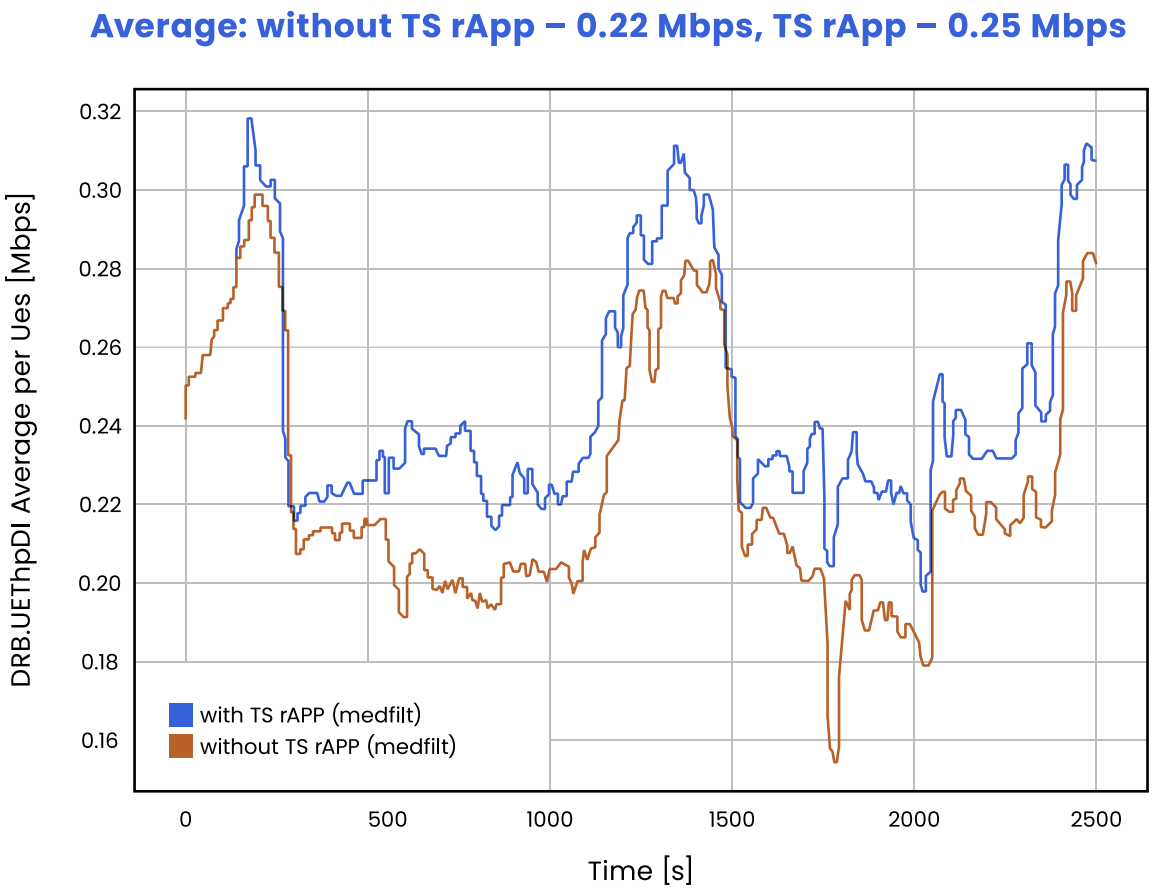


Fig. 3.2-2. Average UEs DL throughput (with TS-rApp: blue, without TS-rApp: orange)

The simulation results presented in Fig. 3.2-2 show that in DL, around 13% improvement in average UE throughput can be observed, mostly for the cell-edge and median users. Most importantly, during the whole simulation, despite selecting not the strongest cells for some of the users, the achieved throughput is high compared to the baseline without TS-rApp. This leads to higher spectrum efficiency.

This closes the discussion on the TS migration from xApp in Near-RT RIC to rApp in SMO/Non-RT RIC, and proves that this concept applies to real networks.

4.0 Beamforming Optimization Use Cases

An advanced use case defined by the O-RAN ALLIANCE in [15], highlights the relevance of adaptive beamforming and M-MIMO optimization, along with its applicability to the SMO/Non-RT RIC framework. These use cases call for rApps dedicated to adjusting beam parameters such as azimuth and elevation, based on real-time network KPIs or information from external sources, e.g., location information. This is applicable for a GoB approach to M-MIMO, which relies on a predefined set of beams that are launched in fixed directions. The main challenge of GoB is that the fixed set of beams may not track the dynamics of a real network, like users' mobility.

Before going to the actual discussions on the algorithm itself, in this chapter we discuss four different use cases for GoB optimization in SMO/Non-RT RIC. They range from static to dynamic beam adaptation.

4.1 GoB Use Cases - Introduction

The GoB optimization can be realized in different time scales, ranging from static BO during deployment, through once-per-week or per-month adjustments, to dynamic BO based on location information, e.g., adapted every minute.

Moreover, optimization goals can vary from coverage optimization or reduction of inter-beam ping-pongs to improvement of UE throughput. To illustrate this diversity, in the remaining part of this chapter, we identified four BO use cases.

4.2 Static/Default GoB Configuration

The first use case is called *Static GoB* optimization, as shown in Fig. 4.2-1, where the default GoB set during deployment is adjusted to maximize the coverage. This scenario is closely related to network planning. Therefore, the most beneficial GoB configuration can be determined using advanced simulators, e.g., based on ray-tracing tools. The key PMs considered in this use case, aiming at coverage maximization, are RSRP and SINR. The *Static GoB Configuration* is sufficient for environments with predictable coverage needs, such as rural sectors or areas with little variation in user distribution.

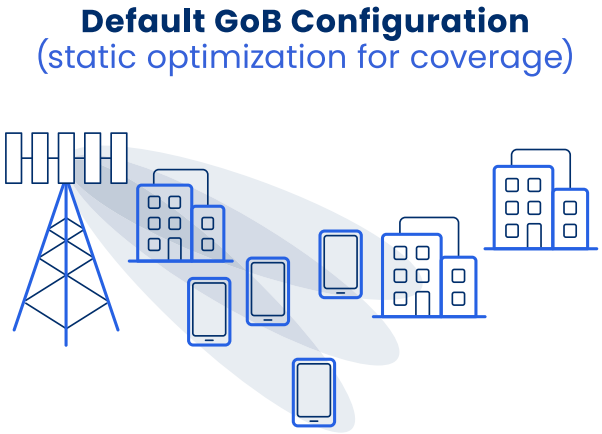


Fig. 4.2-1. Static/Default GoB Configuration

4.3 Event-Based/Semi-Static GoB Configuration

While the fully static approach to GoB optimization may be sufficient for serving day-to-day mobile traffic, it can suffer from poor performance under unnatural network loads. This can be related to, e.g., social events, a concert, or a football match. We refer to this use case as *event-based GoB optimization* (see Fig. 4.3-1). By considering the nature of sport or social events, which are typically planned, this introduces semi-static GoB reconfiguration, where the new GoB is determined by ray-tracing simulations to deal with the expected network traffic increase in a specific place. With that additional information, the static GoB configuration can be changed according to the one optimized for a planned event. After the event ends, the GoB configuration changes can be reverted to the default one. This occasional adjustment of GoB can significantly improve the QoS of a user who takes part in the event. Metrics to evaluate performance are similar to the static GoB optimization, i.e., RSRP and SINR. However, considering additional information about data volume, latency, or cell loads from past events can be beneficial.

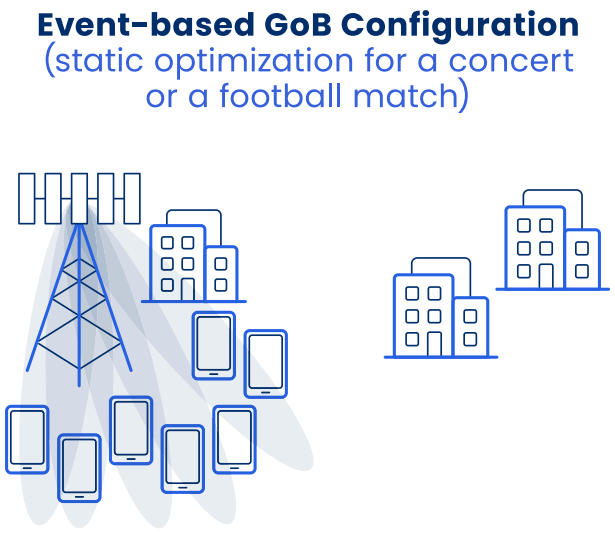


Fig. 4.3-1 Event-Based/Semi-Static GoB Configuration

4.4 Slow Dynamic GoB Configuration

Event-based GoB Configuration is an approach to BO optimization that considers some aspect of dynamic network conditions. One step further is to collect network data and observe the long-term daily traffic patterns to adapt the GoB configuration to the actual throughput demand. This enables a closed-loop optimization that continuously reacts to changes in the network state. We refer to this use case as Slow Dynamic, depicted in Fig. 4.4-1. It approaches the situation with a periodic KPI-driven optimization, where the system regularly evaluates PMs and adjusts the GoB orientation to maximize a chosen objective, e.g., maximizing UE's overall throughput. The PMs to be considered in this use case are the distribution in time of RSRP and SINR. The timescale of the adjustment depends on long-scale dependencies in the network, e.g., morning-afternoon or weekday-weekend, when the variations are significant. To capture dependencies, the optimization algorithm needs to analyze UEs' throughput at each time of day or day of the week. When such variations are noticed, the regular adjustment of the GoB may be done. To this end, location information can play an important role. Even having information about the general routes of UEs over different times of the day, GoB can be configured to point in those directions.

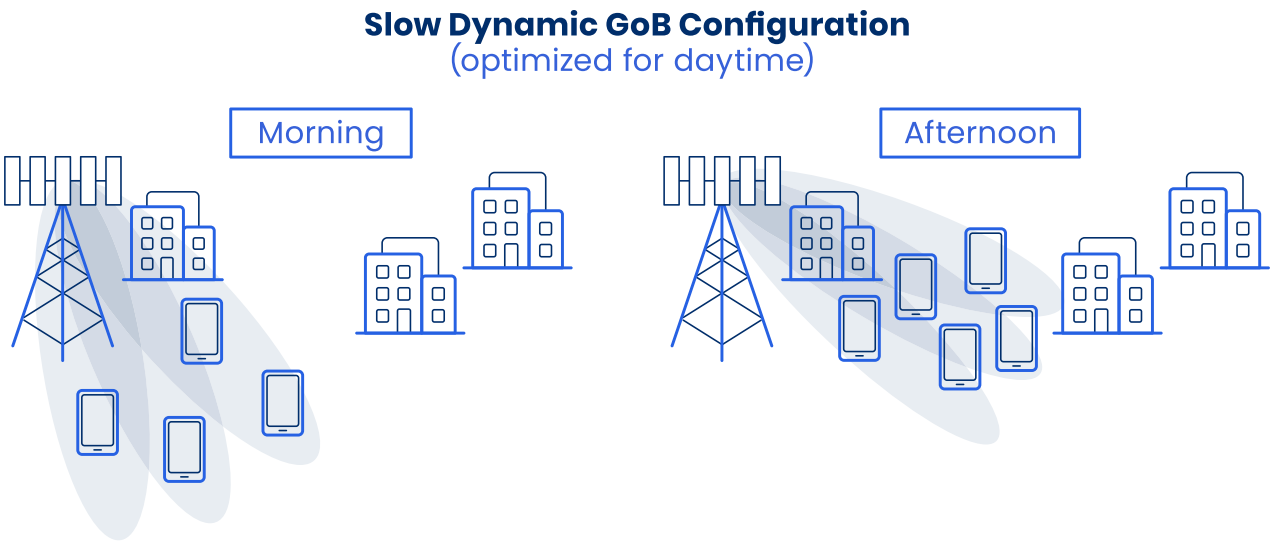


Fig. 4.4-1 Slow Dynamic GoB Configuration

4.5 Fast Dynamic GoB Configuration

Finally, the most dynamic scenario, where the beams are adjusted regularly at relatively small intervals, is presented in Fig. 4.5-1, i.e, *fast dynamic GoB configuration*. In this case, the frequency of decisions on GoB configuration depends on, e.g., hardware limitations for the O-RAN Radio Unit (O-RU), or interface limitations in terms of signaling. Typically, it is 5-15 minute intervals in practical deployments. For this purpose, the network needs the UE's location information provided, e.g., from external data sources or predictions generated by an NDT. This can be augmented by SINR or RSRP measurements in time distribution, as well as radio resource utilization per beam. Beam orientation can be adapted proactively by anticipating upcoming fluctuations in traffic distribution or users' movements. Compared to the previous use cases, *Fast Dynamic GoB Configuration* represents the most advanced form of GoB optimization. It is a continuously adjusting, intelligence-driven approach that reflects both present and future network conditions, potentially creating an opportunity to benefit from utilizing advanced AI/ML algorithms.

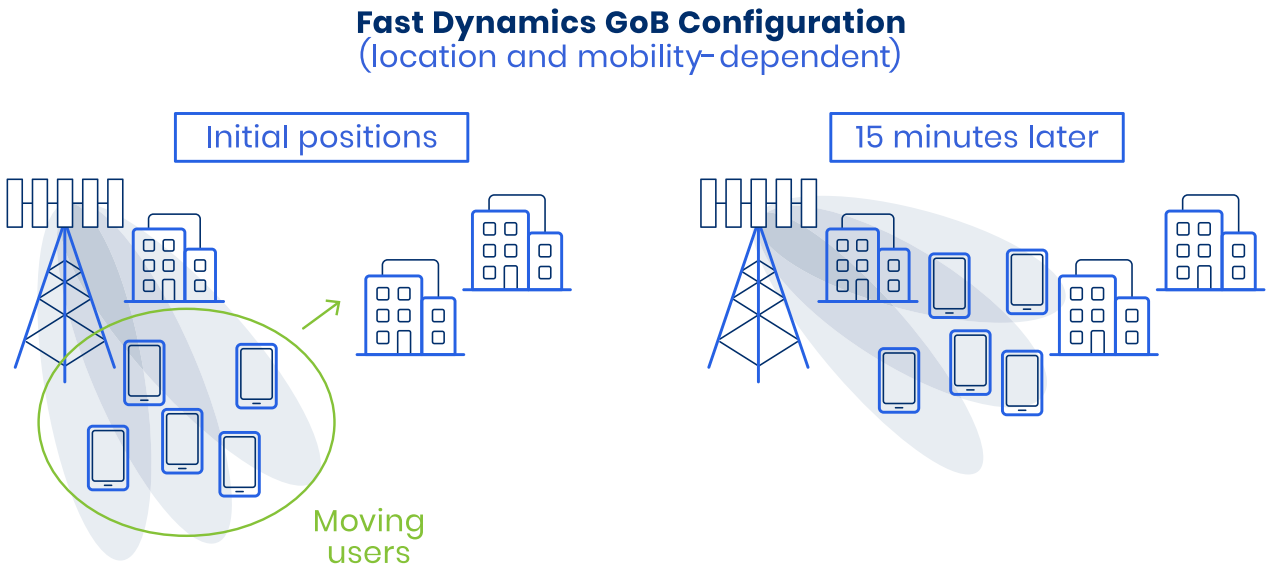


Fig. 4.5-1 Fast Dynamic GoB Configuration with Moving Users

Taking the above discussions into account, in the next chapter, we discuss the actual design of the BO-rApp developed by Rimedo Labs. It addresses the GoB optimization use cases presented throughout Chapter 4.0 and their applicability to the O-RAN architecture.

5.0 Beamforming Optimization-rApp: Design & Results

Beamforming optimization algorithm can be realized as an rApp addressing GoB optimization use cases discussed in Chapter 4.0. Designing of BO-rApp requires careful alignment of the input and output data with technical specifications from both O-RAN ALLIANCE and 3GPP. This would ensure its applicability to the live networks.

In this chapter, we discuss the design of the BO-rApp within the O-RAN architecture, including potential use of NDT or location information. Chapter 5.0 is finalized with numerical results demonstrating the benefits of the BO-rApp targeting Event-based GoB configuration.

5.1 BO-rApp in the O-RAN Architecture

As already mentioned in previous chapters, in the current O-RAN automation market, there is a shift towards SMO/Non-RT RIC, reflected by growing interest in automation platforms. This was our motivation to implement TS features as an rApp in Chapter 3.0. Similarly, we propose the BO-rApp for the Non-RT RIC with the decision-making logic for GoB adjustments (see Fig. 5.1-1). Its role is to analyze performance trends based on the PMs obtained over the O1 interface, integrate available external data, like location information, when needed, and determine the optimal orientation for the entire beam set. These decisions are translated into actionable configuration updates sent through the O1 interface to the O-DU or O-RU [16], depending on the implementation.

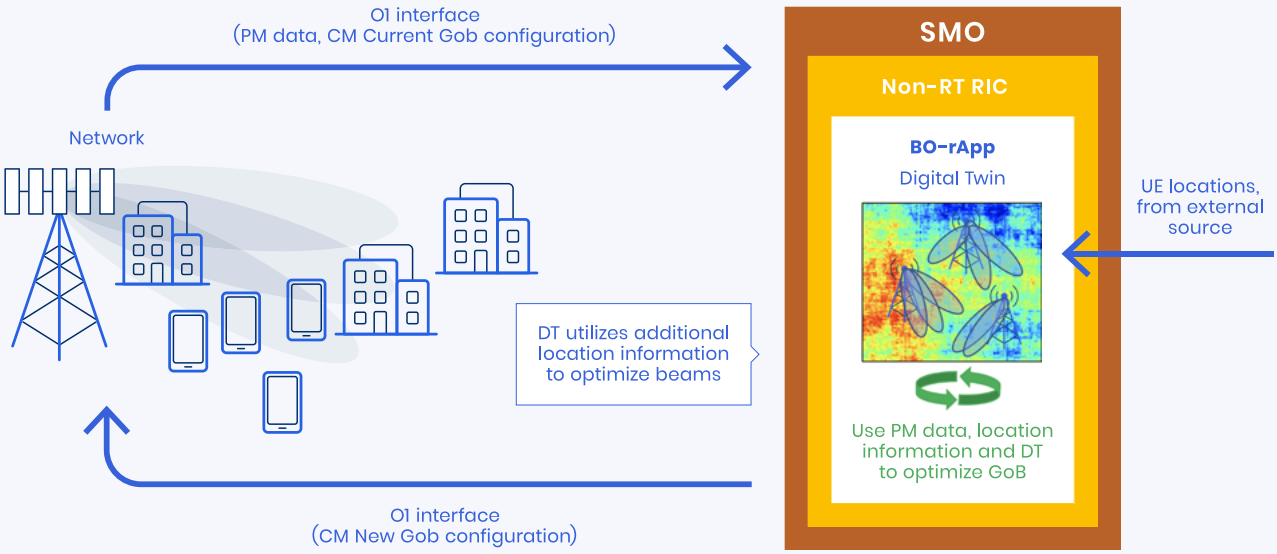


Fig. 5.1-1 Simplified BO-rApp deployment into O-RAN architecture

The BO-rApp monitors network PMs such as RSRP and SINR distributions, as well as traffic load. Depending on the use case, BO-rApp can adjust GoB configuration dynamically or statically, depending on the operator's needs, available data, and RAN/radio capabilities. When available, BO-rApp can incorporate additional sources of information, such as UE location (useful for fast dynamic GoB optimization – discussed in Chapter 4.5), or operator-provided policies. The BO-rApp computes the azimuth and elevation shifts for the GoB and applies the changes via the O1 interface. BO-rApp can also be equipped with NDT, which allows it to test the proposed decisions (GoB configurations) by executing them inside NDT and choosing the most promising configuration to be applied in the network. The advantage here is that the NDT does not operate in the actual real-world network, thus being more secure than applying the algorithm directly into real-life configuration.

While GoB optimization in a network, with hundreds of M-MIMO BSs, it can become a complex problem. Therefore, the BO-rApp can make use of Agentic AI or Reinforcement Learning solutions to find the optimal solution. AI Agents can be deployed within the NDT and work on the synthetic data to propose initial configurations. However, it is crucial to incorporate feedback from the live network to assess their actions in the real world.

The BO-rApp utilizes the O1 interface to modify the GoB configuration; thus, this process is strictly related to the underlying 3GPP specifications. Specifically, TS 28.541 [10] describes the relevant CM and ensures that the BO-rApp complies with the standardized beamforming management. The key metrics that can be considered for BO-rApp are:

» Common Beamforming Function: digitalAzimuth, digitalTilt;

» Beam: beamIndex, beamType, beamAzimuth, beamTilt, beamHorizWidth, beamVertWidth;

» Cell, carrier, sector parameters like: bandwidth, ARFCN UL/DL, location, antenna parameters.

Additionally, the PMs observed by the BO-rApp over the O1 interface to adjust the GoB configuration, from 3GPP TS 28.552 [7], are:

» SS-SINR distribution per SSB and per gNB;

» SS-RSRP distribution per SSB and per gNB;

» SS-RSRQ distribution per SSB and per gNB.

5.2 BO-rApp Utilization for Event-Based GoB Deployment

Let us now discuss the performance gains related to the BO-rApp. Suppose we have a statically configured M-MIMO network, and that there is an ongoing event involving hundreds of UEs located in a specific area. This fits the *Event-Based GoB optimization* use case discussed in Chapter 4.3. We selected it to demonstrate the operation of the BO-rApp. To obtain numerical results, we conducted a series of simulations with 1000 UEs located near the M-MIMO BS, as presented in Fig. 5.2-1. The rApp uses NDT to examine a few options of GoB configuration and selects the best one associated with the highest UE-received SINR, which, in this case, is a 15-degree rotation.

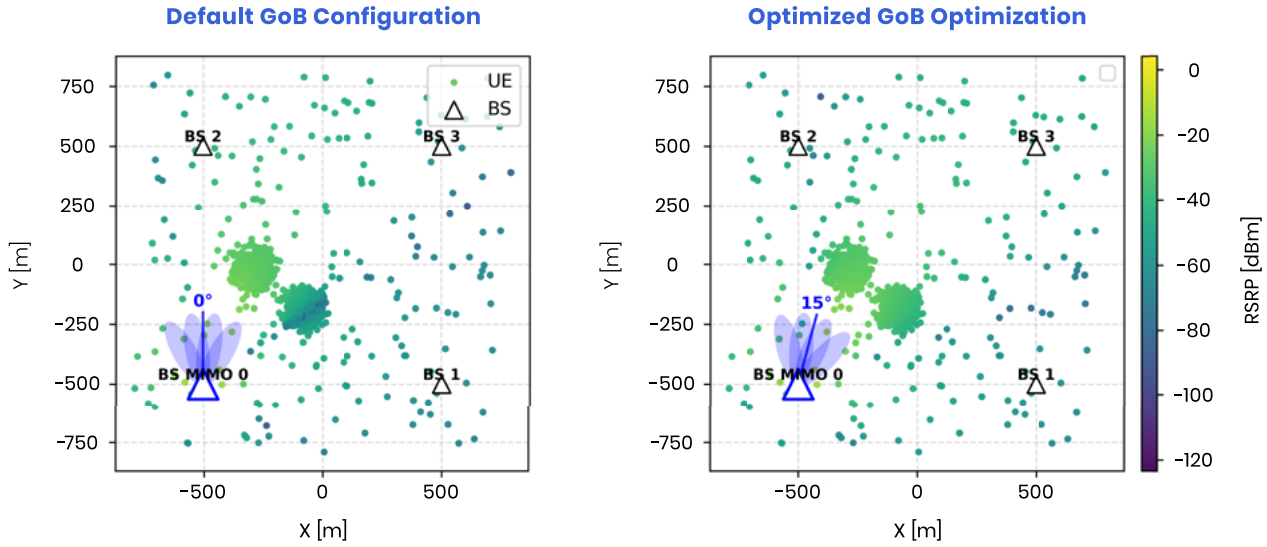


Fig. 5.2-1. RSRP in GoB configuration, Default vs. Optimized

Fig. 5.2-2 and Fig. 5.2-3 provide the comparison between the SINR and RSRP distributions observed for the default GoB configuration (pointing towards 0 degrees) and the GoB configuration optimized by the BO-rApp for the event's location. A clear improvement is seen when the GoB is rotated by 15 degrees. The median SINR increases by approximately 2 dB, indicating a significantly better alignment between the beam directions and the dominant UE cluster. An even bigger gain is in RSRP, where the improvement is up to 10 dB.

As a result, the highly directional characteristic of M-MIMO beams is beneficial in this scenario. By rotating the GoB, the rApp effectively recalibrates the radiated power toward the UE groups participating in the event, and thus improves both RSRP and SINR.

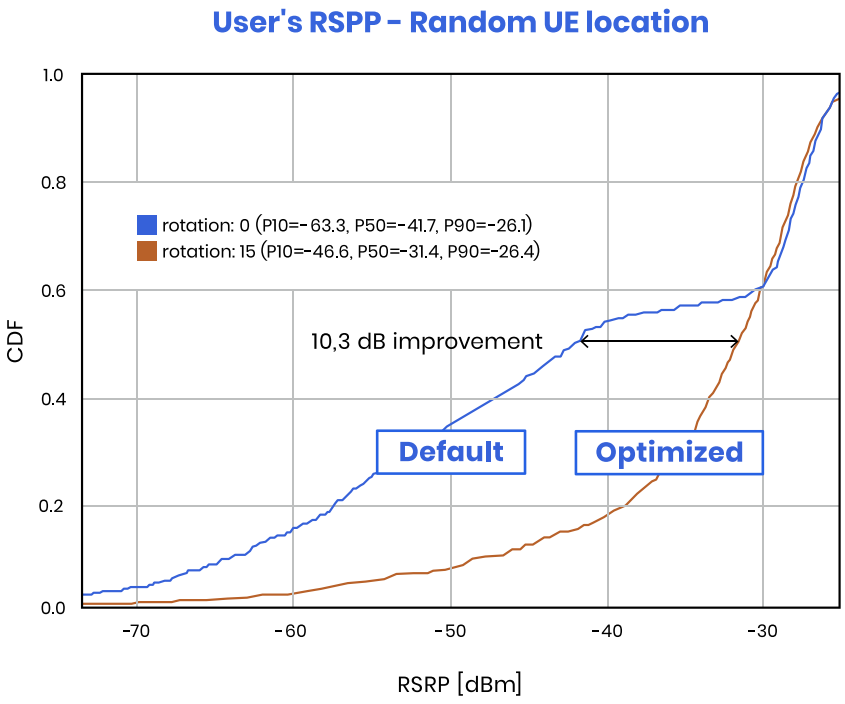


Fig. 5.2-2. Cumulative Distribution Function of user's RSRP in random location scenario (Default: blue; Optimized: orange)

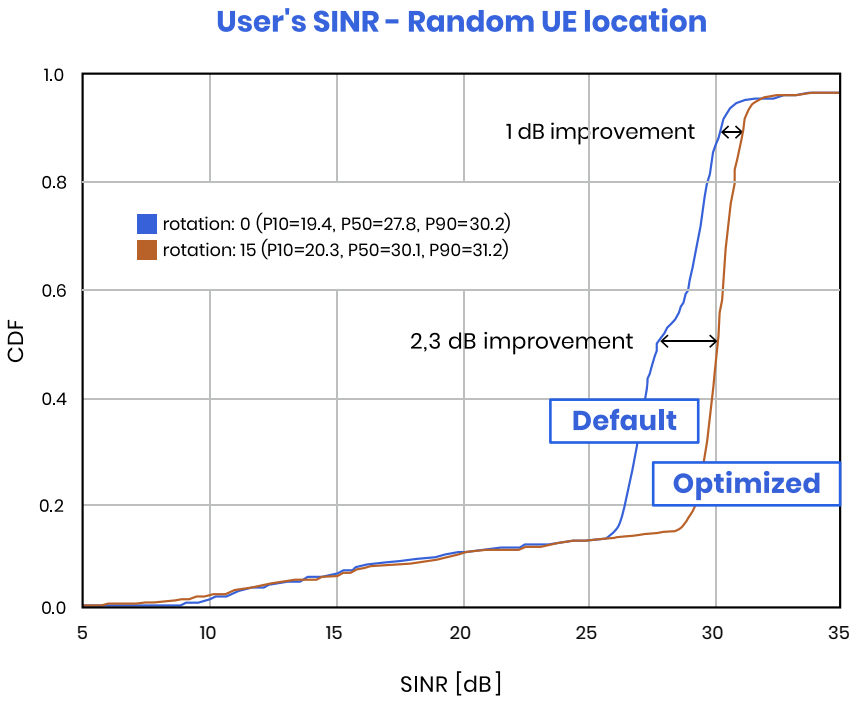


Fig. 5.2-3. Cumulative Distribution Function of user's SINR in random location scenario (Default: blue; Optimized: orange)

Rimedo Labs continues the development of BO-rApp and its integration with SMO platforms. We expect that addressing more challenging Slow/Fast GoB configuration use cases (see Chapter 4.0) will provide higher gains.

Summary & Conclusions

Currently, to meet MNOs' demand for intelligent RAN automation, the Open RAN market is adapting the SMO/Non-RT RIC framework. The leading NEPs announce their automation platforms, enabling development of third-party RAN optimization algorithms in the form of rApps. This has two consequences: first, new solutions to the RAN automation should be designed to fit the SMO/Non-RT RIC framework; second, existing solutions originally designed for Near-RT RIC are being rethought and migrated to rApps. The whitepaper covers these aspects, starting from introducing SMO/Non-RT RIC architecture, followed by detailed discussions on TS-rApp and BO-rApp along with their design, implementation, and results.

In Chapter 1.0, the SMO/Non-RT RIC functional architecture is introduced, and complemented by an overview of the rApps concept, with two examples: TS-rApp and BO-rApp. It serves as a first step, providing basic information about those, leading towards more detailed discussions in the remaining part.

One of the flagship solutions provided by Rimedo Labs is TS-xApp. Due to the wide adoption of SMO/Non-RT RIC, it was brought up in the form of an rApp. To this end, Chapter 2.0 provides a comparison between Near-RT and Non-RT RIC frameworks in the context of TS. In particular, the E2 interface and xApp deployment allow for direct and fast control over user-to-cell association. In contrast, the TS-rApp encompasses a long-term LB alternating CIO through the O1 interface.

Chapter 3.0 continues the discussion on the TS feature in SMO/Non-RT RIC. In the SMaRT-5G project, TS-rApp has been integrated with the OSC-Non RT RIC and subsequently validated using a commercial-grade RAN emulator. The TS-rApp demonstrated a throughput improvement of approximately 13% when evaluated in a setup with 17 cells and 80 emulated UEs. The performance boost was primarily noticeable for cell-edge and median users, who experienced the biggest gains. TS-rApp intelligently redistributes traffic, even directing some users to cells with slightly worse radio conditions but more available radio resources, maximizing overall network efficiency.

Chapter 4.0 touches another RRM aspect of 5G networks, namely, GoB optimization targeting M-MIMO BSs. We indicate its feasibility for the SMO/Non-RT RIC framework, and identify 4 practical GoB optimization use cases: static, semi-static, slow dynamic, and fast dynamic. While static BO requires the simplest metrics, such as RSRP and SINR, it mostly fits the fixed scenarios. On the other hand, the fast dynamic BO yields higher throughput improvements, but requires frequent PM reporting and the UE's updated location information.

Finally, in Chapter 5.0, we propose a BO-rApp fitting the SMO/Non-RT RIC framework according to O-RAN ALLIANCE specifications. The BO-rApp uses ray-tracing-based NDT for the evaluation of possible GoB configurations. The proposed concept was validated by simulations of an event-based semi-static configuration showing RSRP and SINR improvements.

The overall conclusions from this whitepaper are as follows:

- The Open RAN automation market is shifting toward SMO/Non-RT RIC, making rApps a natural choice for deploying third-party AI/ML algorithms in 5G networks.
- To fit the SMO/Non-RT RIC framework, some of the solutions in the form of xApps are being redesigned.
- While designing new rApps, O-RAN ALLIANCE and 3GPP specifications related to the O1 interface must be taken into account.
- While a more natural TS environment is Near-RT RIC enabling per-UE data collection and handover control, some features like LB can be realized in SMO/Non-RT RIC.
- While the O1 interface has some limitations, the TS-rApp delivers tangible improvements in throughput and user performance under realistic network scenarios.
- Rimedo Labs, together with other members of the SMaRT-5G initiative, demonstrated how to adopt traffic steering to rApp and achieve significant QoS gains.
- BO is one of the advanced RRM topics targeting 5G M-MIMO networks, addressable by rApps. For the GoB optimization, we identified 4 use cases:
  - Static – related to the network planning, and intended to be optimized for the average day-to-day traffic;
  - Event-based/semi-static – introducing adjustment of the GoB to the planned sport or social event;
  - Closed-loop slow dynamic – extending the previous use cases to track network patterns associated with a day (morning, afternoon) and adjust GoB accordingly;
  - Closed-loop fast dynamic – with GoB configuration being adjusted to the current UE mobility.
- The BO, or more specifically, GoB adjustment, can be realized within the O-RAN framework relying on the configuration structures and PMs defined by 3GPP.
- Rimedo Labs' BO-rApp is a practical solution enabling intelligent BO in automated M-MIMO networks. It bridges the gap between static GoB deployments and fully adaptive beamforming by introducing AI/ML and using ray-tracing-based NDT for a safe evaluation of possible GoB configurations.

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Glossary

5G	The 5th Generation of Mobile Networks
5QI	5G QoS Indicator
6G	The 6th Generation of Mobile Networks
AI	Artificial Intelligence
API	Application Programming Interface
BO	Beamforming Optimization
BS	Base Station
CCC	Cell Configuration and Control
CIO	Cell Individual Offset
CM	Configuration Management
EI	Enrichment Information
gNB	gNodeB
GoB	Grid-of-Beams
IOC	Information Object Class
KPI	Key Performance Indicator
KPM	Key Performance Measurement
LB	Load Balancing
ML	Machine Learning
M-MIMO	Massive MIMO
MNO	Mobile Network Operator
NDT	Network Digital Twin
Near-RT RIC	Near-Real Time RIC
NEP	Network Equipment Provider
Non-RT RIC	Non-Real Time RIC
O-RAN	Open RAN (by O-RAN ALLIANCE)
O-RU	O-RAN Radio Unit
OSC	O-RAN Software Community
PM	Performance Management
PRB	Physical Resource Block
QoS	Quality of Service
RAN	Radio Access Network
rApp	Application to run on Non-RT RIC
RC	RAN Control
RIC	RAN Intelligent Controller
RRM	Radio Resource Management
RSRP	Reference Signal Receive Power
SA	Standalone
SBA	Service-Based Architecture
SINR	Signal-to-Interference-and-Noise Ratio
SM	Service Model
SMO	Service Management and Orchestration
TE&IV	Topology Exposure and Inventory
TM	Traffic Management
TS	Traffic Steering
UE	User Equipment
V2X	Vehicular-to-Anything
xApp	Application to run on Near-RT RIC

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Marcin Hoffmann is a Technical Solution Manager at Rimedo Labs, working on O-RAN software development solutions and R&D projects covering energy savings, traffic steering, and massive MIMO. Marcin is a Graduate Student Member at IEEE and received the M.Sc. degree (Hons.) in electronics and telecommunication from Poznań University of Technology in 2019, where he is currently pursuing a Ph.D. degree with the Institute of Radiocommunications. He was involved in many national and international research projects. His research interests include the utilization of machine learning and location-dependent information for network management. He coauthored many scientific articles published in top journals like IEEE Journal on Selected Areas in Communications, IEEE Transactions on Intelligent Transportation Systems, IEEE Communications Magazine, and IEEE Access.



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Marcin Dryjanski received his Ph.D. (with distinction) from the Poznan University of Technology in September 2019. Over the past 15 years, Marcin served as an R&D engineer and consultant, technical trainer, technical leader, advisor, and board member. Marcin has been involved in 5G design since 2012 when he was a work-package leader in the FP7 5GNOW project. Since 2018, he has been a Senior IEEE Member. He is a co-author of many articles on 5G and LTE-Advanced Pro and a co-author of the book „From LTE to LTE-Advanced Pro and 5G" (M. Rahnema, M. Dryjanski, Artech House 2017). From October 2014 to October 2017, he was an external advisor at Huawei Technologies Sweden AB, working on algorithms and architecture of the RAN network for LTE-Advanced Pro and 5G systems. Marcin is a co-founder of Grandmetric, where he served as a board member and wireless architect between 2015 and 2020. Currently, he serves as CEO and principal consultant at Rimedo Labs.

About Rimedo Labs

Rimedo Labs specializes in providing high-quality consulting, implementation, and R&D services in the field of modern wireless systems, focusing on Open RAN, AI-RAN, 5G, and beyond. Rimedo Labs is a software provider delivering customized Apps for RAN optimization.

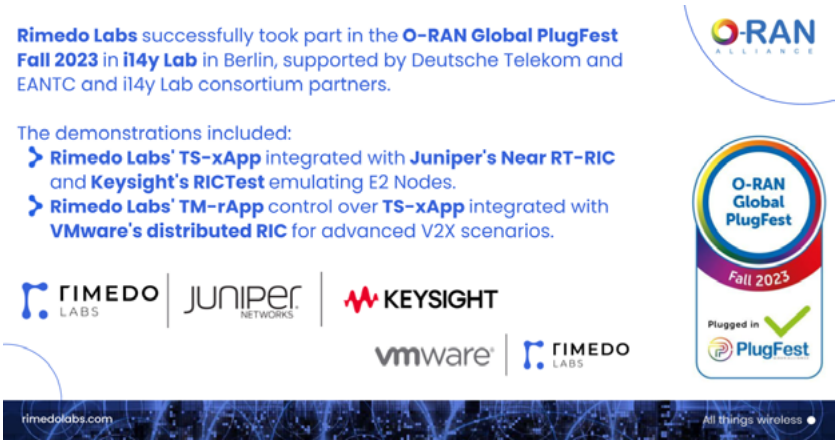
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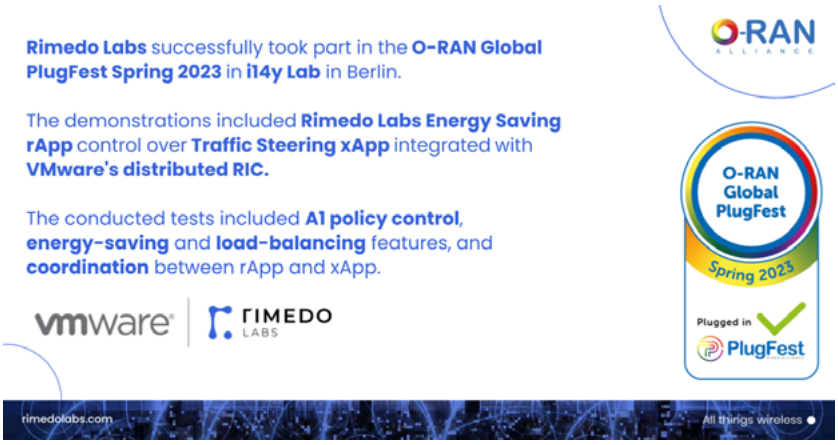
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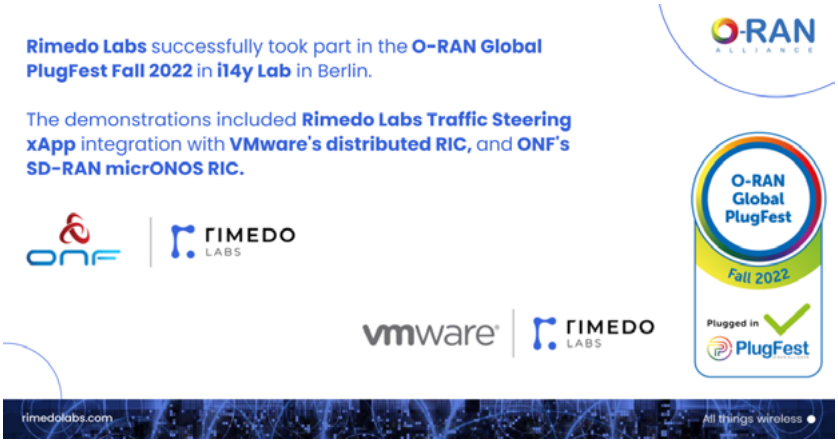
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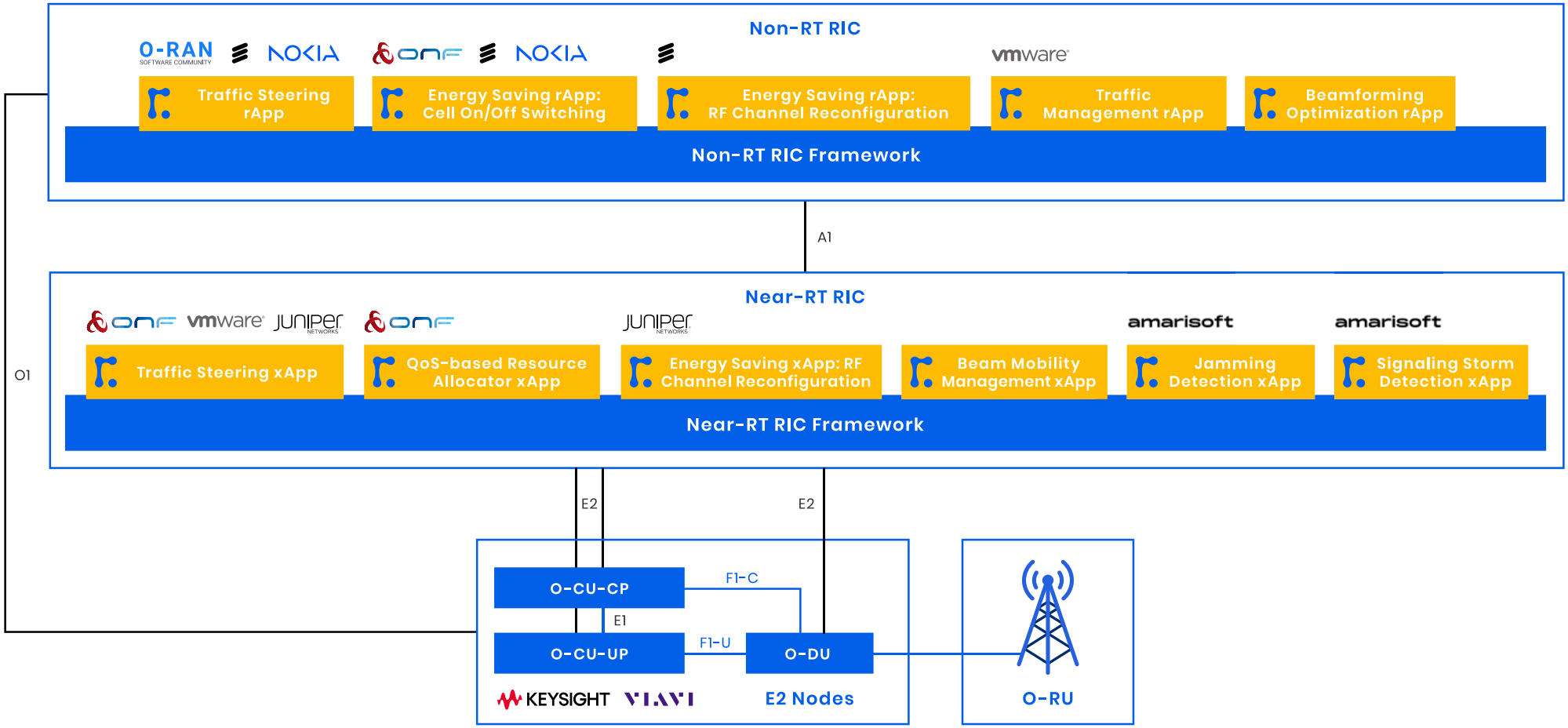








Rimedo Labs xApp/rApp Portfolio



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